

A Logistic Regression Approach in Predicting Employee Attrition for a Data-Driven Support System in Human Resources

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Abstract—Employee turnover presents a significant challenge to organizational stability and financial performance. This study develops a predictive model utilizing Logistic Regression to identify key drivers of attrition within an employee dataset of 3,100 observations. Through systematic data preparation and feature selection, the research identifies age, salary, and bonuses as primary numerical indicators of resignation. Experimental results demonstrate that a baseline model with a balanced regularization parameter achieves an accuracy of 68.79%, successfully balancing model complexity and predictive power. The findings suggest that while numerical models provide a foundational early-warning system, their real-world value is maximized when integrated into a broader Human Resources framework that addresses unquantifiable personal factors through proactive talent management interventions.

Keywords—Logistic Regression, Employee Attrition, Predictive Analytics, Human Resources, Hyperparameter Tuning, Decision Support System

I. INTRODUCTION

The utilization of predictive algorithms to analyze datasets is a fundamental approach in data analytics. A key distinction among the modeling techniques relates to whether the target objective is categorical or numerical in nature. Linear Regression is specialized in predicting continuous numerical values [1], accordingly it is not suitable, and designed for classification problems where the result is binary, such as "True" or "False". Moreover, Linear Regression follows a straight path, wherein it can predict values that are less than 0 or greater than 1, which is the main reason it is not logical for it to be used in classification. In contrast, Logistic Regression ensures that the model output is always interpreted as a probability between 0 and 1, making it the viable option for binary classification. This specific approach of using Logistic Regression combined with feature selection has been proven effective in identifying the primary drivers of employee attrition [2].

In order to achieve a probability-based output, Logistic Regression encompasses the Sigmoid Function. A mathematical component responsible in mapping any real-valued input into a range strictly between 0 and 1. The mapping is necessary because raw linear predictions can theoretically be any number from negative to positive infinity, which exceeds the logical limits of probability.

Finally, Logistic Regression utilizes log-odds to connect the data to the probability of the outcome. The log-odds, also known as the logit, is the natural logarithm of the ratio

between success and failure. This step is essential because it allows the model to calculate a linear decision boundary even when dealing with probabilities. While a default threshold of 0.5 is usually used to assign categories, this limit can be adjusted to avoid critical errors, such as missing a positive result in a medical test.

II. METHODOLOGY

In this section, the systematic methodology for constructing the predictive model is detailed, from data acquisition to the final preparations for Logistic Regression.

A. Dataset Description

The research utilizes an employee record collection titled *HR_Dataset_by_Akel.xlsx*, which was obtained from an HR analytics repository. The dataset contains 3,100 observations, representing individual employee profiles, across 11 distinct variables. These variables capture a wide range of information, including demographic data such as *Age* and *Gender*, professional details like *Department* and *Job Role*, and financial metrics including *Salary*, *Bonus*, and *Total Compensation*.

B. Data Preparation

To ensure high data quality, several cleaning steps were performed to remove noise and resolve inconsistencies. The identifiers *EmployeeID* and *Name* were removed because they contain unique labels that do not provide generalizable predictive value. Additionally, *TotalCompensation* was excluded to avoid redundant information, as it is a direct sum of other existing features like *Salary* and *Bonus*. The target variable, *Attrition*, was standardized by converting entries to lowercase and removing extra spaces before being mapped to a binary numeric format. To handle numerical anomalies, impossible values such as ages outside the 18–60 range or negative salaries were neutralized. These missing values, along with outliers in the *Bonus* category, were replaced using the column median to maintain a logical distribution and ensure the model was not skewed by extreme values.

C. Feature Selection

```
--- Pearson Correlation Analysis ---
Attrition    1.000000
Age          -0.220838
Salary       -0.275762
Bonus        -0.282539
Name: Attrition, dtype: float64
```

Figure 1. Pearson Correlation Analysis

The modeling process used a specific set of independent variables chosen because they relate to why employees leave, specifically Age, Salary, and Bonus. To check these choices, the Pearson correlation coefficient was used to measure the relationship between each predictor and the binary target variable, Attrition, where a value of 1 means an employee left and 0 means they stayed. The Pearson test results showed that all three variables have a negative relationship with attrition, with Salary and Bonus showing the strongest links. This means that as pay and age increase, the chance of an employee leaving the company decreases. By focusing on these specific predictors and their correlation results, the model identifies the main numerical drivers behind employee resignation.

D. Train-Test Split

The data underwent structural transformations to ensure mathematical compatibility with the Logistic Regression algorithm. Numerical features were scaled using a MinMaxScaler to bring all variables onto a consistent 0 to 1 range. For the final experimental setup, the refined dataset of 3,060 records was partitioned using a standard hold-out method. Following an 80/20 ratio, 2,448 observations were allocated to the training set to allow the model to identify predictive relationships. The remaining 612 observations were reserved as the testing set to verify the model's predictive accuracy on unseen data. This partitioning is essential to validate that the system can generalize its findings to new employee profiles.

III. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, the performance of the Logistic Regression model is evaluated through different configurations.

A. Effects of C-Value on Complexity and Accuracy

The regularization parameter, or C-value, was adjusted to observe its impact on the model's ability to generalize. When a very small C-value of 0.001 was applied, the model became too simple and failed to identify patterns, resulting in a low accuracy of 49.18%. In this state, the model predicted that no employees would leave, showing that high regularization leads to underfitting. Conversely, the baseline model with a C-value of 1.0 achieved a significantly higher accuracy of 68.79%. This suggests that a balanced C-value allows the model enough complexity to capture the relationships between compensation, age, and attrition without becoming overly sensitive to noise.

B. Feature Influence and Performance Changes

Based on the coefficients of the Logistic Regression model, salary and bonus were identified as the features with the strongest influence on prediction. The negative coefficients for these variables indicate that as financial incentives increase, the probability of an employee leaving the company decreases. During the experiments, focusing only on these numerical drivers allowed the model to maintain a stable performance. While categorical features like Job Role were initially considered, the numerical metrics provided the most consistent signal for identifying at-risk employees.

C. Trade-off between False Positives and False Negatives

The experiment revealed a critical trade-off between False Positives (predicting an employee will leave when they actually stay) and False Negatives (predicting an employee will stay when they actually leave). With the current settings, the model achieved a balanced precision and recall of approximately 0.69 for both classes. Reducing False Negatives is often prioritized in HR settings because missing an employee who is about to quit is more costly than mistakenly identifying a loyal employee for a retention interview. However, increasing sensitivity to capture more resignations usually leads to more False Positives, which could waste company resources on unnecessary retention efforts.

D. Threshold Justification

For this study, the standard classification threshold of 0.5 was used to separate employees likely to stay from those likely to leave. However, in a professional HR environment, a lower threshold may be more effective. If the company wants to prevent losing talent, lowering the threshold helps identify more at-risk employees. While this might lead to more False Positives, it is often better to offer a retention talk to a loyal employee than to miss someone who is actually planning to resign. Therefore, the current model provides a strong data-driven foundation for a support system aimed at improving employee retention and workforce stability.

E. Threshold Justification

TABLE I. MODEL PERFORMANCE COMPARISON ACROSS VALIDATION METHODS

Metric	Baseline (Test-Split)	5-Fold CV	10-Fold CV
Accuracy	69.00%	67.97% (Mean)	67.78% (Mean)
Stability (Std Dev)	N/A (Single Run)	2.03%	2.68%
Best Performance	69.00%	70.26%	71.57%
Worst Performance	69.00%	64.38%	63.73%

The experimental results shown in Table I., demonstrate a clear trade-off between optimistic performance and statistical reliability across the three evaluation methods. The Baseline Test-Split yielded the highest single accuracy of 69.00%, yet this proved to be a slightly "lucky" estimation when compared to the more rigorous K-Fold methods. Implementing 5-Fold Cross-Validation provided a more realistic mean accuracy of 67.97% with high stability (2.03% standard deviation), while the 10-Fold Cross-Validation offered the most exhaustive "stress test," resulting in a mean of 67.78% and a wider variance of 2.68%. Collectively, the transition from a single split to 10-fold validation reveals that while the model is mathematically stable, its true predictive

power sits closer to 68% than the initial 69%, with the 10-fold approach exposing specific data sensitivities that a simple test-split fails to capture.

IV. PERFORMANCE EVALUATION

In this section, the best-performing model, which is the baseline Logistic Regression with a C-value of 1.0, is evaluated. This model provided the most reliable results by balancing complexity and accuracy during the experiments.

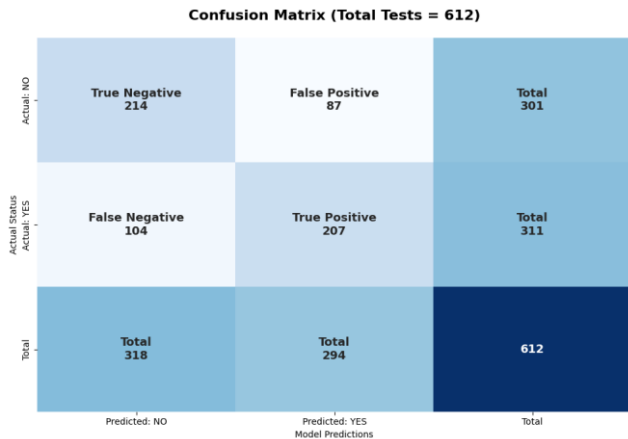


Figure 2. Pearson Correlation Analysis

The confusion matrix, shown in Figure 2. is used to visualize the performance of the model by comparing the actual values against the predicted outcomes. Based on the test set of 612 observations, the model correctly identified 214 employees who stayed (TN) and 207 employees who left (TP). However, there were 87 cases where the model incorrectly predicted an employee would leave (FP) and 104 cases where it failed to identify an employee who actually resigned (FN). This matrix shows that the model is effective at identifying both classes but has a slightly higher rate of missing actual resignations.

TABLE II. MONITORING TABLE FOR HYPERPARAMETER TUNING EXPERIMENT

Experiment	C Value	Accuracy	Precision	Recall	F1 Score	Observation
Baseline Model	1.0	68.79 %	0.69	0.69	0.69	Balanced performance with stable generalization.
Strong Regularization	0.001	49.18 %	0.24	0.49	0.32	Underfitting occurred because the model was too simple to find patterns.

Weak Regularization	1000	68.95 %	0.69	0.69	0.69	High complexity with minimal gain and potential for overfitting.
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The performance of the model is analyzed by comparing the metrics from three different hyperparameter tuning experiments shown on Table II. The Baseline Model achieved an accuracy of 68.79% and an F1-score of 0.69, which shows a balanced performance. In contrast, the Strong Regularization experiment resulted in a much lower accuracy of 49.18% and an F1-score of 0.32 because the model was too simple to find patterns in the data. The Weak Regularization experiment showed a slightly higher accuracy of 68.95% and an F1-score of 0.69. However, the Baseline Model remains the preferred choice because it provides a similar performance with lower complexity, which reduces the risk of overfitting. These results confirm that the selected parameters provide a stable and reliable model for HR decision-making.

V. CONCLUSION

The development of this predictive model provides a data-driven perspective on employee turnover, demonstrating that financial incentives like salary and bonuses, alongside age, may serve as primary indicators for resignations. While the baseline model achieved an accuracy of 68.79% and benefited from a balanced C-value of 1.0 to ensure stability, its real-world value lies in its role within a larger Human Resources framework. The system only analyzes numerical data and cannot account for unquantifiable personal factors such as family situations, mental well-being, or workplace culture; therefore, HR professionals must use these predictive scores as a starting point for qualitative interventions like stay-interviews and wellness checks. This technical approach allows a professional HR department to move from reacting to resignations, to preventing them through proactive talent management. By combining technical accuracy with professional human interaction, management can address individual needs before a resignation occurs, ultimately improving workforce stability by treating employees as people rather than just numbers.

REFERENCES

- [1] McCarthy, Mary & Ceccucci, Wendy. (2022). Applying Predictive Analytics: Finding Value in Data. [10.1007/978-3-030-83070-0](https://doi.org/10.1007/978-3-030-83070-0).
- [2] Wardhani, Fitri & Lhaksmana, Kemas. (2022). Predicting Employee Attrition Using Logistic Regression With Feature Selection. Sinkron. 7. 2214-2222. [10.33395/sinkron.v7i4.11783](https://doi.org/10.33395/sinkron.v7i4.11783).